



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2025– 2026 (Odd Semester)

Degree, Semester & Branch : VII Semester B.E CSE
Course Code & Title : AI3021 & IT in Agricultural System
Name of the Faculty member : Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit V/E-GOVERNANCE IN AGRICULTURAL SYSTEMS/Decision Support System
- **Course Outcome:**5
- **Topic Learning Outcome:** 42
- **Activity Chosen:** Case Study Analysis
- **Justification:**
 - DSS in agriculture involves real-time decisions, and case studies provide realistic scenarios for students to practice decision-making.
 - Complex concepts like crop growth models, soil analysis tools, weather-based advisory systems, and yield prediction become easier to understand through practical situations.
 - Students develop analytical, problem-solving, and decision-making skills by evaluating actual agricultural data and selecting the best DSS solution.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - The instructor explains the purpose of DSS in agriculture such as crop planning, resource allocation, pest prediction, and climate risk management.
 - Students receive a case scenario (e.g., a farmer facing drought, low soil fertility, or pest attack).
 - Data includes rainfall, soil nutrients, crop type, weather forecasts, and available technologies.
 - Students work in small groups to analyze the case.
 - Each group discusses the problem, identifies constraints, and explores DSS tools that can support the farmer.
 - Students apply knowledge of agricultural DSS modules such as:
 - Crop simulation models (DSSAT)
 - Soil moisture estimation tools
 - Weather-based forecasting models
 - Farm management decision advisors

- Irrigation scheduling tools
- Each group develops a data-driven decision plan (e.g., irrigation schedule, fertilizer recommendation, crop selection).
- Groups present their solution to the class.
- Other students critique, question, and offer alternative strategies—promoting deeper understanding.
- The instructor clarifies misconceptions, highlights important DSS components, and connects student solutions to real agricultural practices.
- Students reflect on what they learned about decision-making, agricultural data interpretation, and the importance of DSS in modern farming.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO5	PO8	PO10	PO12	PSO3
CO4	3	3	3	3	3	3	3

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO4	PO5	PO10	PO12	PSO2
	3	3	3	1	2	2	2
Justification for correlation	Students able to	Students able to	Students able to	Students able to	Students able to	Students able to	Students able to

• **Images / Screenshot of the practice:**



Fig 1: Student discussed the topics Figure2:Sivasundari presented ideas

Reflective Critique:

❖ Feedback from Students and Stakeholders:

- Students reported that the case study analysis helped them clearly understand how Decision Support Systems (DSS) are used in real agricultural decision-making. They felt that working with realistic farm data such as rainfall predictions, soil nutrient levels, and irrigation schedules made the learning more practical and meaningful.
- Many students shared that analyzing the farmer's situation helped them connect theoretical DSS models with real-world farming challenges like irregular rainfall, crop selection, and pest management.
- Students appreciated the group-based approach, as it allowed them to discuss multiple solution strategies, compare different crop choices, and understand how seasonal forecasts influence farm decisions.

❖ Benefits of the Practice:

- Promotes real-world problem-solving.
- Enhances students' ability to interpret agricultural data and model outputs.
- Strengthens teamwork, communication, and critical analysis.
- Connects classroom concepts with practical agricultural applications.
- Improves understanding of technology-based farming solutions.

❖ Challenges Faced:

- Some students may struggle with interpreting agricultural datasets.
- Limited exposure to real DSS tools may create difficulty in applying concepts.
- Unequal participation among team members during case discussion.
- Requires additional time for students to explore and understand multiple decision alternatives.

References:

- ❖ https://en.wikipedia.org/wiki/Peer_learning
- ❖ [Peer to Peer Learning - Examples, Benefits & Strategies](#)
- ❖ <https://whatfix.com/blog/peer-to-peer-learning/>

Signature of Faculty Member

HOD